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UNIVERSITY OF MARYLAND
GLOBAL CAMPUS

Unit 4 Lab Student Report
Collection and Examination
of Digital Evidence

Table of Contents

Instructions 2

Introduction 3

Screenshot 1 – FAT32 Image 4

Screenshot 2 – FAT32 Image 5

Screenshot 3 – FAT32 Image 6

Screenshot 4 – NTFS Image 7

Screenshot 5 – NTFS Image 8

Screenshot 6 – NTFS Image 9

Conclusion..... 10

APA References..... 11

Introduction

Unit 4's lab focused on creating case files in Autopsy and comparing the MD5 and SHA1 hashes to what was taken with FTK Imager in an earlier unit. The process was simple. We opened Autopsy, created a new case file, pointed it to both the FAT32 image and NTFS image on separate cases and let Autopsy ingest the information with pre-configured default modules. After Autopsy finished ingesting the data, the MD5 and SHA1 hashes were compared to what was taken with FTK Imager. They all matched showing that the integrity of the images was maintained.

Once the images were ingested into Autopsy, the user can click around to find a variety of data depending on the modules used. All of the files are visible, along with a pie chart depicting percentages of files being images, deleted, or unknown. Because the images there were run were just of folders made, there is not much data to sort through. What is interesting, is the default files that the appropriate file system use can be found. This is helpful to know as it can help determine what file system is being used. In this case, it is either FAT32 or NTFS.

Screenshot 1 – FAT32 Image

Add the screenshot of your “Autopsy Optional Information” window. The use of anyone else’s work will result in an academic integrity review by your professor. Label your screenshot to receive full credit.

Figure 1: Results of Autopsy Optional Information

New Case Information

Steps

1. Case Information
2. **Optional Information**

Optional Information

Case

Number: Lab 4-1

Examiner

Name: Robert Simpson

Phone:

Email:

Notes: FAT32 Image

Organization

Organization analysis is being done for: Not Specified

< Back Next > Finish Cancel Help

Screenshot 2 – FAT32 Image

Add the screenshot of your “Autopsy Summary” window showing the hash values. Label your screenshot

to receive full credit.

Figure 2: Results of Summary tab of FAT32 Drive.001 and hash values at bottom

Listing

/img_FAT32 Drive.001

Table Thumbnail Summary

Types User Activity Analysis Recent Files Past Cases Geolocation Timeline Ingest History Container

Display Name: FAT32 Drive.001

Name: FAT32 Drive.001

Device ID: b52e71a0-c5ba-4ff0-ab76-1eb75ace88ae

Time Zone: Etc/UTC

Acquisition Details:

Image Type: Raw Single

Size: 1.07 GB (1070596096 bytes)

Unallocated Space: 1.06 GB (1057095680 bytes)

Sector Size: 512 bytes

MD5: 8cc0e35fbb1a6633736369e09fef5e83

SHA1: c0a9d660bb90c4ecbca38a60ac8051f0bc42cdd2

SHA256: d5e3033d070ee031844e5f3c97f3d2b145bcca44907247553fdb31bc201b12a8

File Paths: H:\FAT32 Image\FAT32 Drive.001

Screenshot 3 – FAT32 Image

Add the screenshot of your “FAT32 Drive.001.txt” file (hash log of your FAT32 image). Please label your screenshot to receive full credit.

Figure 3: Hash values in FAT32 Drive.001.txt match the hash values in Autopsy summary

FAT32 Drive.001.txt - Notepad

File Edit Format View Help

Created By AccessData® FTK® Imager 4.7.2.11

Case Information:

Acquired using: ADI4.7.2.11

Case Number: Lab 2-FAT32

Evidence Number: 2A

Unique description: G:Logical Drive

Examiner: Robert Simpson

Notes: UMGC VM

Information for H:\FAT32 Image\FAT32 Drive:

Physical Evidentiary Item (Source) Information:

[Device Info]

Source Type: Logical

[Drive Geometry]

Bytes per Sector: 512

Sector Count: 2,091,008

[Physical Drive Information]

Removable drive: False

Source data size: 1021 MB

Sector count: 2091008

[Computed Hashes]

MD5 checksum: 8cc0e35fbb1a6633736369e09fef5e83

SHA1 checksum: c0a9d660bb90c4ecbca38a60ac8051f0bc42cdd2

Screenshot 4 – NTFS Image

Add the screenshot of your “Autopsy Optional Information” window. Label your screenshot to receive full credit.

Figure 4: NTFS Image Optional information window in Autopsy

New Case Information

Steps

1. Case Information
2. **Optional Information**

Optional Information

Case

Number: Lab 4-1

Examiner

Name: Robert Simpson

Phone:

Email:

Notes: NTFS Image

Organization

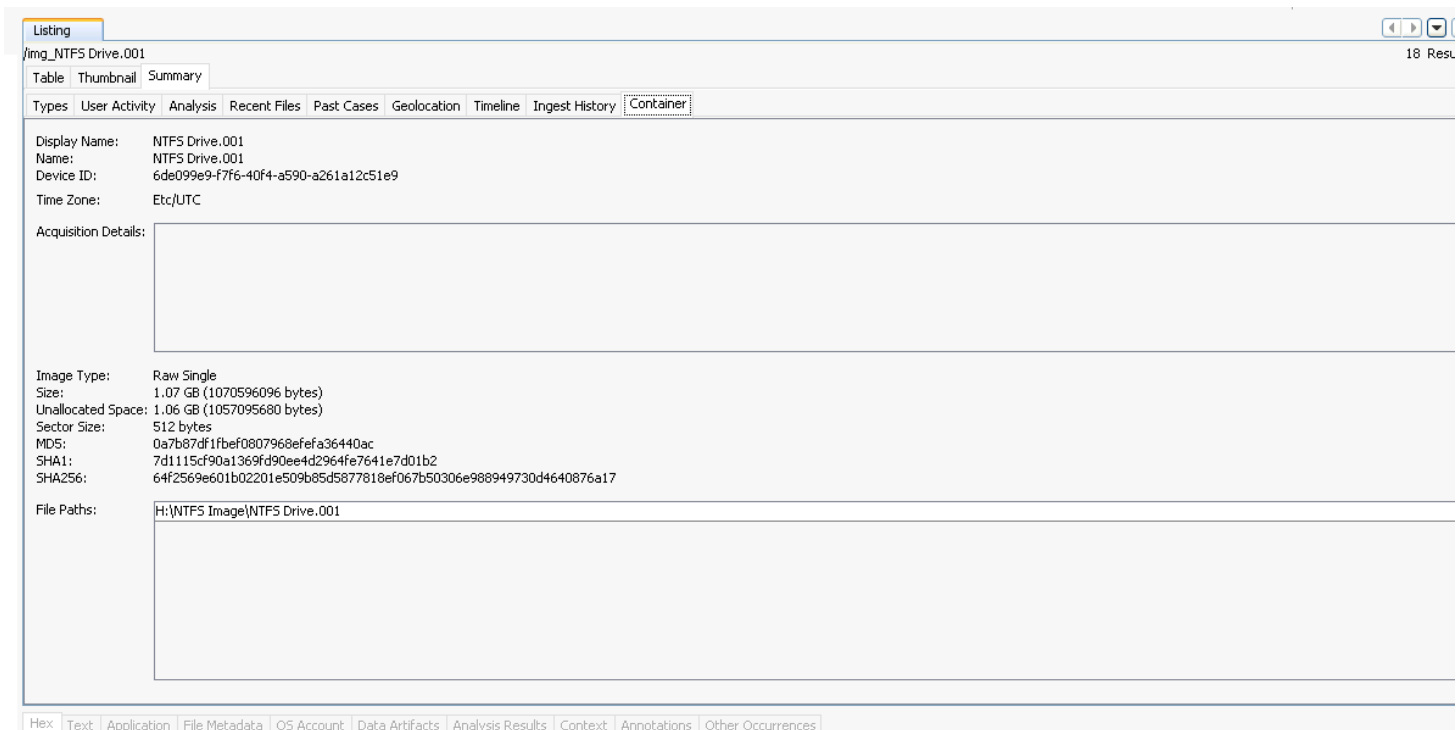
Organization analysis is being done for: Not Specified Manage Organizations

< Back Next > Finish Cancel Help

Screenshot 5 – NTFS Image

Add the screenshot of your “Autopsy Summary” window showing the hash values. Label your screenshot to receive full credit.

Figure 5: Result of Autopsy Summary on NTFS Drive.001 with hashes at bottom



Screenshot 6 – NTFS Image

Add the screenshot of your “NTFS Drive.001.txt” file (hash log of your NTFS image). Please label your screenshot to receive full credit.

Figure 6: Results of matching NTFS Drive.001 hash found in NTFS Drive.001.txt

Image Verification Results:

Verification started: Mon Mar 9 18:25:19 2026

Verification finished: Mon Mar 9 18:25:24 2026

MD5 checksum: 0a7b87df1fbef0807968efefa36440ac : verified

SHA1 checksum: 7d1115cf90a1369fd90ee4d2964fe7641e7d01b2 : verified

Conclusion

Overall, this lab clarified the basics of creating a case file and ingesting data into Autopsy.

Because the images used did not contain the data that would typically be found in an actual case, not much further can be found. However, the lab provided the essential guidance on how to get started with Autopsy and using proper naming conventions. It also emphasized the importance of comparing hash results between programs to ensure nothing impacted the integrity of the image.

The readings supporting unit 4's lab highlight just how important solid digital forensic analyst work is. From the readings provided and other books and podcasts I have consumed surrounding cybersecurity; forensic analysts have an extremely difficult job. Many analysts, especially those in law enforcement, end up dealing with child pornography cases. Because the individuals involved try their best to cover up their tracks and erase their presence, having skilled analysts is crucial to convicting the right people. It is not an easy job, but one that needs to be done.

The readings also show just how new the field of digital forensics is and how it will constantly evolve. Just as blue team SOC analysts fight evolving attacks on networks, forensic analysts must continuously learn new skills to combat new techniques to erase data and tracks. As mentioned in the reading, Operation Innocent Images started in 1994 when chatrooms were a major form of communication (FBI History, 2022). In 2026, analysts have to deal with cryptocurrency transactions, encrypted communications, and new techniques daily. The task of hunting down criminals gets harder as technology progresses.

Encase, FTK Imager, and Autopsy are just some of the tools used in the field. As new tools are created, analysts must learn them on the job. Certifications, courses, and education systems struggle to stay up-to-date with new technologies and tools. They provide a great foundation, but it is on the professional to continue their own education by conducting research and testing new tools. Part of this can be achieved by being involved in forums and attending conferences.

References

FBI History. (2022, May 3). *Operation innocent images*. Federal Bureau of Investigation.

<https://www.fbi.gov/history/famous-cases/operation-innocent-images>